(a) The collapse regime, where rank works - and where we did NOT falsify it

rank value and co-measured collapse evidence

clean in-batch InfoNCE, no queue
diagnostic-script centred effective rank (the diagnostic's own definition; NOT asserted equal to R1, R2 or R3 here)

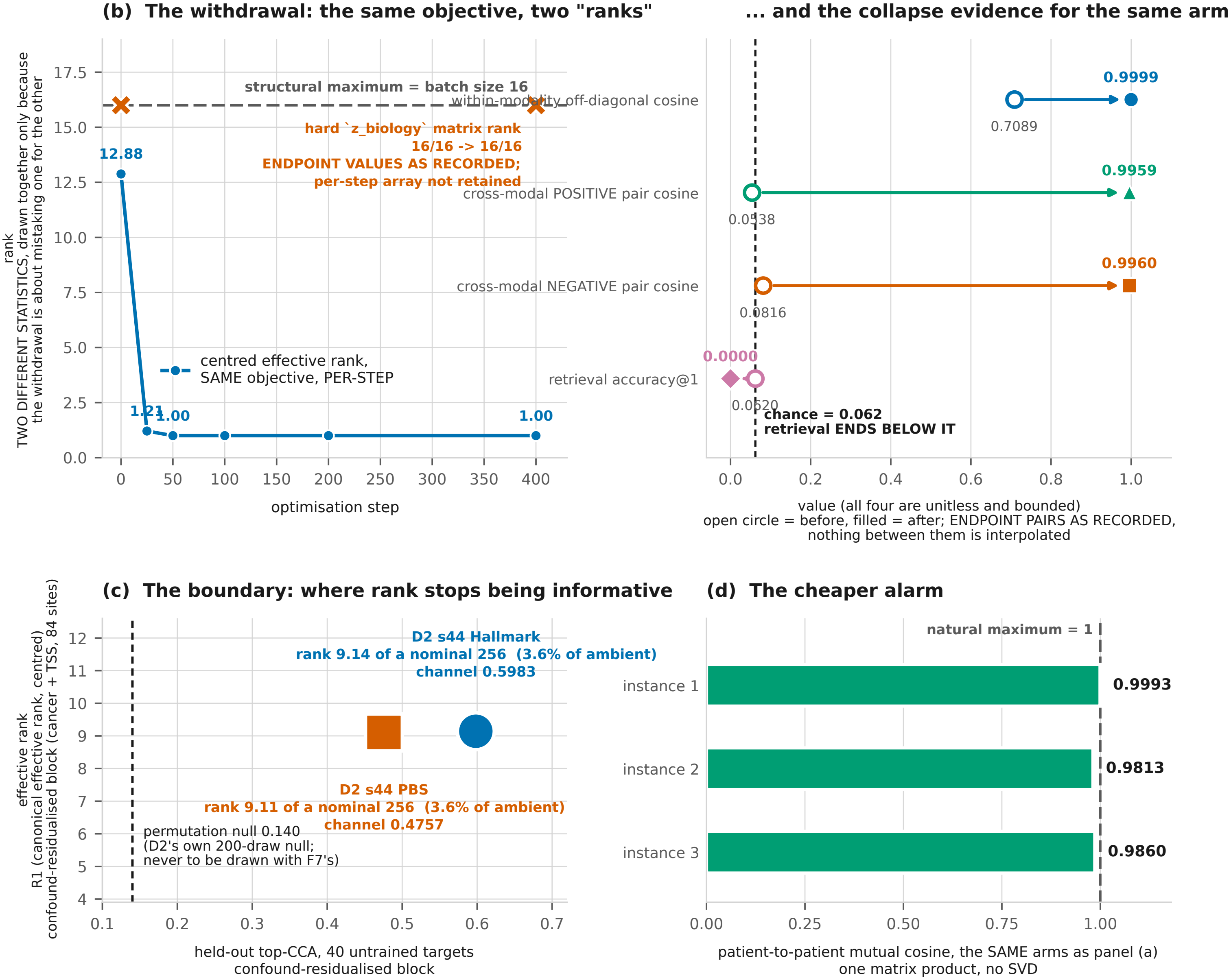
12.88 -> 1.00 by step 50
positive / worst-negative cosine 0.9993 / 0.9993
minimum margin -0.219 -> -0.0001

five programme_free configurations, one common initialisation
R3 (order-2 Hill, row-normalised)
centred, fixed 256-patient held-out probe

67.55 -> about 2 by step 150 (1.86-2.50 across five arms)
RNA-view mutual cosine 0.9813 at step 200
no loss weighting prevents it, including both terms at zero

programme_free seed 42, real D1 training, 282 held-out patients
R3 (order-2 Hill, row-normalised)
held-out probe on the live checkpoints

1.76 at epoch 23 and 1.71 at epoch 39
RNA-RNA mutual cosine 0.977 / 0.986
hard rank 9 / 11 of 256



Two artifacts whose rank agrees to within the sampling noise of the estimator (9.14 against 9.11; F4(a) puts the gap at 1.4 sampling sd) and whose channels differ by 0.1226. At 3.6% of ambient dimensionality, rank is uninformative about the channel in BOTH directions.

0.981-0.9993 in every case, saturating at a natural maximum of 1. This is why the draft recommends the cosine over rank even for the one use of rank that survives.

F8. THE WITHDRAWAL, in full, because without (iv) and (v) this figure must not be published. (i) The 16/16 column is a HARD NUMERICAL RANK (torch.linalg.matrix_rank), not R1, R2 or R3, and a 16 x 256 float matrix has full row rank under essentially any perturbation; (ii) its maximum is 16 because the batch is 16; (iii) it is a TRAIN batch of 16, not held out; (iv) the CENTRED EFFECTIVE RANK OF THE SAME OBJECTIVE FALLS TO 1.00 by step 50, so this instance is evidence FOR the collapse-diagnostic use, not against it; (v) this project previously listed it among its two strongest instances and THAT DESCRIPTION IS WITHDRAWN, here and in P1. Cross-modal POSITIVE and NEGATIVE pairs end indistinguishable, 0.9959 against 0.9960 - the negatives marginally HIGHER - and in-batch InfoNCE ends at 2.7734 against its own chance of 2.7726. Panel (b)'s 16/16 track and all four collapse-evidence quantities are ENDPOINT PAIRS as recorded in ~/e0_run/collapse_diag.log, which probes before and after rather than on a schedule; no per-step array exists and none is interpolated. The source script for the original 16/16 measurement is a scratchpad file on the A100 and is not in this repository. The centred effective-rank track IS per-step and is drawn as one. Panel (a) reads "we did not falsify the collapse-diagnostic use", never "we verified it": we have not found a case of total collapse that effective rank missed, and the figure for that absence would imply a symmetry the data do not support. Panel (c)'s null is D2's own 200-draw null and must never share an axis with F7's within-cancer 0.145-0.147; they differ in n, in component count and in the permutation procedure.